

SPECIAL MATRICES FOR QUANTUM COMPUTING

Hermitian, unitary, normal, positive, projector, diagonal, block, and density-matrix structures

A detailed reference linking matrix structure to quantum gates, observables, measurements, state geometry, and time evolution.

Topic 6 in the Quantum Mathematics Foundation series

How to use this volume

This volume collects the matrix classes that appear repeatedly in quantum computing. The aim is not to memorize isolated definitions, but to recognize what each structure guarantees: real measurement values, reversible evolution, orthogonal decomposition, nonnegative probabilities, or efficient block structure.

Prerequisites: complex numbers, vectors, matrices, adjoints, inner products, eigenvalues, and basic bra-ket notation.

Learning objectives

- Distinguish transpose, complex conjugate, and adjoint.
- Classify matrices as Hermitian, skew-Hermitian, unitary, normal, positive semidefinite, projector, diagonal, permutation, or block structured.
- Explain why gates are unitary and observables are Hermitian.
- Use projectors and positive operators to calculate quantum probabilities.
- Recognize the Pauli matrices and common one-qubit gates by structure.
- Connect matrix exponentials, commutators, controlled operations, and density matrices to the special classes.

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1. Structural map: why special matrices matter

A general complex matrix may stretch some directions, collapse others, change normalization, possess complex eigenvalues, and fail to admit an orthogonal eigenbasis. Quantum theory restricts matrices according to their role.

Special matrix classes used in quantum theory

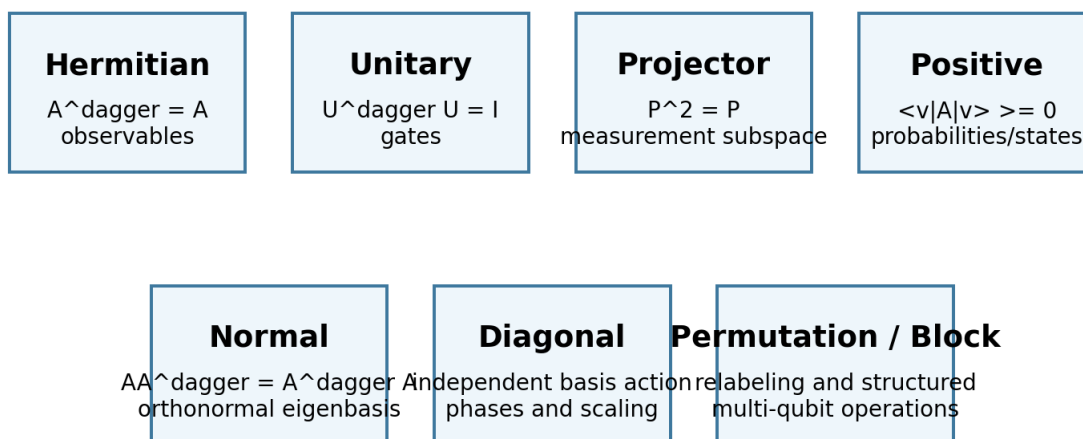


Figure 1. The main special classes used throughout quantum information.

Matrix class	Defining condition	Primary quantum meaning
Hermitian	$A^\dagger = A$	Observable or generator with real eigenvalues
Unitary	$U^\dagger U = I$	Reversible probability-preserving gate
Projector	$P^2 = P$ and usually $P^\dagger = P$	Selects a measurement subspace
Positive semidefinite	$\langle v A v \rangle \geq 0$ for all $ v\rangle$	Nonnegative probabilities and state operators
Normal	$AA^\dagger = A^\dagger A$	Orthonormal eigenbasis
Diagonal	Only diagonal entries may be nonzero	Independent action on basis coordinates
Permutation	One 1 in each row and column	Reversible relabeling of basis states

One matrix may belong to several classes. Pauli Z, for example, is diagonal, Hermitian, unitary, normal, and both its square and inverse equal the identity.

2. Identity, zero, diagonal, triangular, and block matrices

The simplest structures often reveal a quantum operation immediately.

Identity and zero matrices

$$I|\psi\rangle = |\psi\rangle, \quad 0|\psi\rangle = |0_{\text{vector}}\rangle.$$

The identity is unitary, Hermitian, positive definite, and a projector onto the whole space. The zero matrix is Hermitian, positive semidefinite, and a projector onto the zero subspace, but it is not unitary.

Diagonal matrices

$$D = \text{diag}(d_0, d_1, \dots, d_{n-1}).$$

A diagonal matrix multiplies each basis amplitude independently. It is especially important because eigenvalue problems become trivial in the basis where an operator is diagonal.

Input state	$D = \text{diag}(a, b)$	Output
$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$	$\longrightarrow$	$\begin{bmatrix} a\alpha \\ b\beta \end{bmatrix}$

A diagonal matrix acts independently on basis-state amplitudes.

For a unitary diagonal gate, $|a| = |b| = 1$, so only phases change.

Figure 2. Diagonal operators act independently on basis components.

A diagonal matrix is Hermitian exactly when all diagonal entries are real. It is unitary exactly when every diagonal entry has magnitude one.

Triangular and block matrices

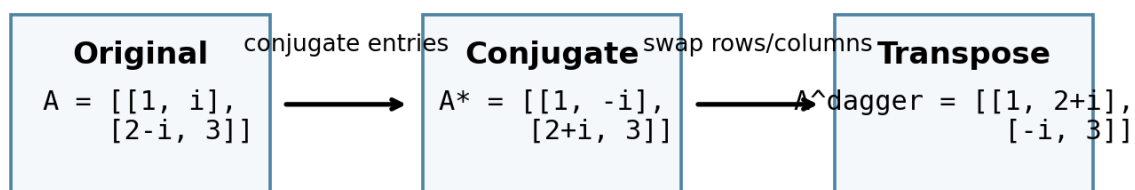
The eigenvalues of an upper- or lower-triangular matrix are its diagonal entries. A block-diagonal matrix acts independently on separate subspaces and has eigenvalues equal to the combined eigenvalues of its blocks.

$$A = \text{diag}(A_1, A_2) \Rightarrow \text{spectrum}(A) = \text{spectrum}(A_1) \cup \text{spectrum}(A_2).$$

3. Transpose, complex conjugate, and adjoint

These three operations must be kept distinct.

Operation	Notation	Action
Transpose	A^T	Swap rows and columns
Entrywise conjugate	A^* or $\overline{A}$	Replace i by $-i$ in every entry
Adjoint / conjugate transpose	$A^\dagger$	Conjugate entries and transpose



The adjoint is the operation required by complex inner products.

Figure 3. The adjoint combines conjugation and transposition.

Essential identities

- $(AB)^\dagger = B^\dagger A^\dagger$: the order reverses.
- $(A^\dagger)^\dagger = A$.
- $(A+B)^\dagger = A^\dagger + B^\dagger$.
- $(cA)^\dagger = c^* A^\dagger$.
- For a ket $|\psi\rangle$, the bra is $\langle\psi| = (|\psi\rangle)^\dagger$.

The adjoint is defined so that inner products satisfy $\langle\phi|A|\psi\rangle = \langle A^\dagger\phi|\psi\rangle$. It is therefore the correct complex-space analogue of the transpose.

4. Hermitian matrices

$$H^\dagger = H$$

A 2 x 2 Hermitian matrix has the form

$$H = \begin{bmatrix} a & z \\ z^* & d \end{bmatrix}, \text{ with } a, d \text{ real.}$$

The diagonal entries must be real, and the off-diagonal entries occur as conjugate pairs.

Why observables are Hermitian

- Every eigenvalue is real, matching real measurement outcomes.
- Eigenvectors belonging to distinct eigenvalues are orthogonal.
- There is an orthonormal eigenbasis in finite dimensions.
- Expectation values $\langle \psi | H | \psi \rangle$ are real.

$$H = \lambda_1 |v_1\rangle\langle v_1| + \lambda_2 |v_2\rangle\langle v_2|$$

real eigenvalues orthogonal projectors onto eigen-directions

Hermitian operators are diagonal in an orthonormal eigenbasis.

Figure 4. Hermitian operators decompose into real eigenvalues times orthogonal projectors.

Spectral decomposition

$$H = \sum_k \lambda_k |v_k\rangle\langle v_k|.$$

This representation says: measure which eigenspace the state occupies, then associate that subspace with the real value λ_k .

Example

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \text{ is Hermitian because } Y^\dagger = Y.$$

Its eigenvalues are +1 and -1, with eigenvectors $(|0\rangle + i|1\rangle)/\sqrt{2}$ and $(|0\rangle - i|1\rangle)/\sqrt{2}$.

5. Skew-Hermitian matrices

$$K^{\dagger} = -K$$

Skew-Hermitian matrices are the complex analogue of real antisymmetric matrices. Their diagonal entries are purely imaginary, and their eigenvalues are purely imaginary or zero.

Relation to Hermitian matrices

$$K \text{ is skew-Hermitian} \iff iK \text{ is Hermitian.}$$

Every matrix can be split uniquely into Hermitian and skew-Hermitian parts:

$$A = (A + A^{\dagger})/2 + (A - A^{\dagger})/2.$$

The first term is Hermitian; the second is skew-Hermitian.

Why this matters for quantum evolution

If H is Hermitian, then $-iH$ is skew-Hermitian. Exponentiating a skew-Hermitian matrix produces a unitary matrix:

$$U(t) = \exp(-iHt) \text{ is unitary.}$$

This is the algebraic reason Hermitian Hamiltonians generate reversible time evolution.

6. Unitary and orthogonal matrices

$$U^\dagger U = UU^\dagger = I$$

A unitary matrix is invertible with

$$U^{-1} = U^\dagger.$$

Unitary transformations preserve the complex inner product:

$$\langle U\phi | U\psi \rangle = \langle \phi | U^\dagger U | \psi \rangle = \langle \phi | \psi \rangle.$$

Consequences

- Norms and total probability are preserved.
- Orthogonal states remain orthogonal.
- Angles and distinguishability are preserved.
- The operation is reversible.
- Every eigenvalue has magnitude one and can be written $\exp(i\theta)$.

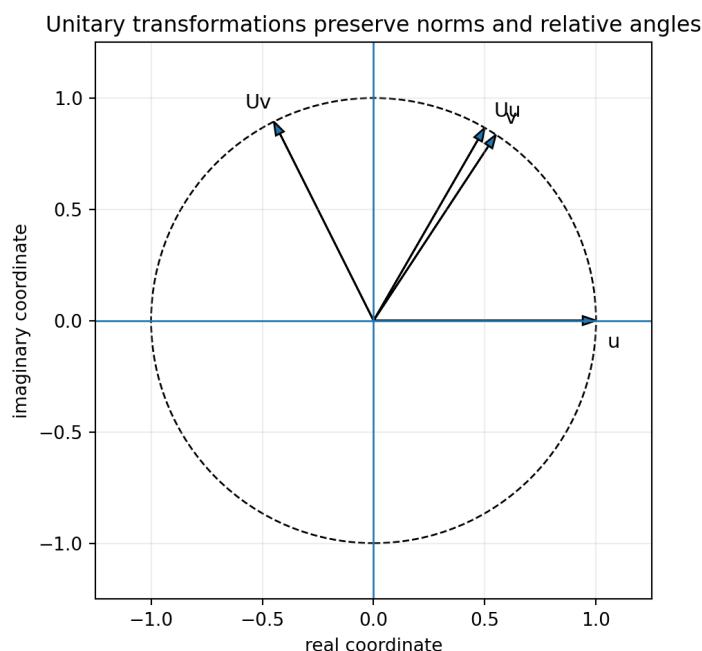


Figure 5. A unitary transformation changes orientation and phase without distorting state-space geometry.

Column test

A square matrix is unitary exactly when its columns form an orthonormal basis. Equivalently, its rows form an orthonormal basis.

Orthogonal matrices

A real unitary matrix is called orthogonal and satisfies $Q^T Q = I$. Plane rotations and reflections are standard examples.

7. Special unitary matrices and global phase

A unitary matrix has determinant of magnitude one:

$$|\det(U)| = 1.$$

The special unitary group $SU(n)$ contains unitary matrices with determinant exactly one:

$$SU(n) = \{U : U^\dagger U = I, \det(U)=1\}.$$

Why determinant phase appears

The determinant is the product of eigenvalues. Since every unitary eigenvalue is a phase, the determinant is also a phase.

Removing global phase

Any U in $U(2)$ can be written as a global phase times an $SU(2)$ matrix:

$$U = \exp(i\gamma) V, \text{ with } \det(V)=1.$$

When U acts on an isolated pure state, the common factor $\exp(i\gamma)$ is a global phase and has no observable effect. This is why one-qubit physical rotations are closely associated with $SU(2)$, though circuit-level controlled operations can make formerly global phases become relative.

Double-cover warning

$SU(2)$ represents rotations of a qubit Bloch vector in a two-to-one way: U and $-U$ produce the same Bloch-sphere rotation because they differ by global phase.

8. Normal matrices and the spectral theorem

$$N N^{\dagger} = N^{\dagger} N$$

Normal matrices are exactly the complex matrices that can be diagonalized by a unitary change of basis:

$$N = V D V^{\dagger}, \text{ where } V \text{ is unitary and } D \text{ is diagonal.}$$

Important subclasses

- Hermitian matrices are normal.
- Unitary matrices are normal.
- Skew-Hermitian matrices are normal.
- Diagonal matrices are normal.
- General matrices need not be normal.

What normality guarantees

- An orthonormal eigenbasis exists.
- Geometric multiplicity equals algebraic multiplicity.
- Functions of the matrix can be computed from its eigenvalues.
- No defective Jordan-block behavior occurs.

Normal does not mean Hermitian

The phase gate $S = \text{diag}(1, i)$ is normal and unitary, but not Hermitian because its eigenvalue i is not real.

9. Projectors and quantum measurement

$$P^2 = P$$

An orthogonal projector additionally satisfies $P^\dagger = P$. For a normalized state $|\psi\rangle$,

$$P|\psi\rangle = |\psi\rangle \langle \psi | \psi \rangle.$$

A projector keeps one component and removes the orthogonal part

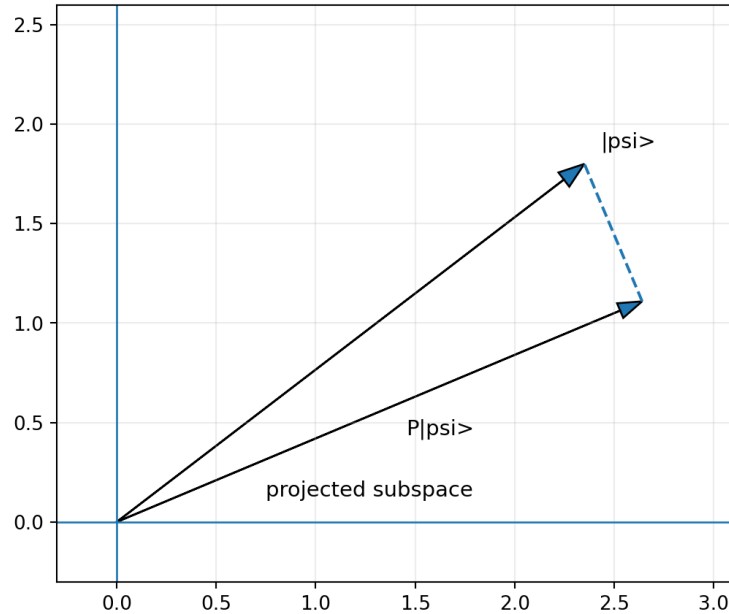


Figure 6. A projector extracts the component parallel to a measurement subspace.

Eigenvalues of a projector

From $P^2 = P$, any eigenvalue obeys $\lambda^2 = \lambda$. Therefore

$$\lambda \in \{0, 1\}.$$

Vectors in the projected subspace have eigenvalue 1; vectors orthogonal to it have eigenvalue 0.

Measurement probability

$$p = \langle \psi | P | \psi \rangle.$$

For $P|\psi\rangle = |\psi\rangle \langle \psi | \psi \rangle$ this becomes the Born rule $p = |\langle \psi | \psi \rangle|^2$.

Complete projective measurement

$$P_j P_k = \delta_{jk} P_k, \sum_k P_k = I.$$

The projectors are mutually exclusive and collectively exhaustive.

10. Positive semidefinite matrices

A Hermitian matrix A is positive semidefinite, written $A \geq 0$, when

$$\langle v | A | v \rangle \geq 0 \text{ for every } |v\rangle.$$

Positive semidefinite quadratic forms never go below zero

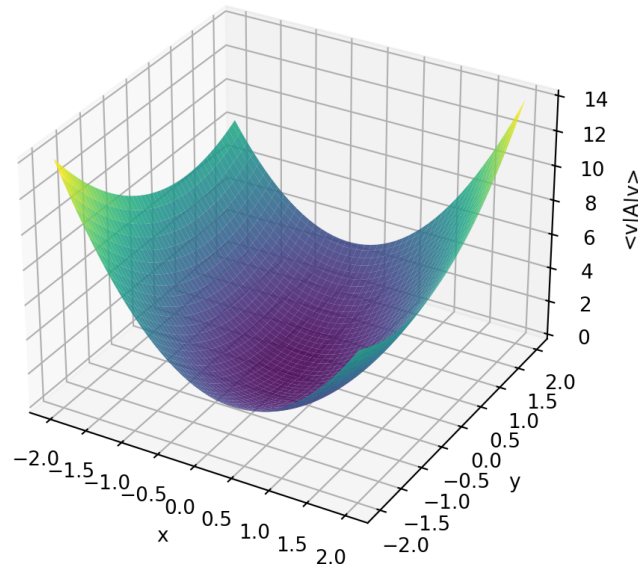


Figure 7. A positive semidefinite quadratic form never takes a negative value.

Equivalent conditions for Hermitian A

- Every eigenvalue is nonnegative.
- $A = B^\dagger B$ for some matrix B .
- A has a unique positive semidefinite square root $\sqrt{A}$.

Positive definite

A is positive definite when $\langle v | A | v \rangle > 0$ for every nonzero vector. Then all eigenvalues are strictly positive and A is invertible.

Quantum relevance

- Projectors are positive semidefinite.
- Density matrices are positive semidefinite.
- POVM measurement effects are positive semidefinite.
- Probabilities written as $\langle \psi | E | \psi \rangle$ are nonnegative because $E \geq 0$.

11. Density matrices as special quantum matrices

A density matrix ρ describes pure states, classical mixtures, and reduced subsystems. It satisfies three core conditions:

$$\rho^\dagger = \rho, \rho \geq 0, \text{Tr}(\rho)=1.$$

Pure state

$$\begin{aligned}\rho &= |\psi\rangle\langle\psi| \\ \rho^2 &= \rho \\ \text{Tr}(\rho^2) &= 1\end{aligned}$$

Mixed state

$$\begin{aligned}\rho &= \sum p_k |\psi_k\rangle\langle\psi_k| \\ \rho^2 &\neq \rho \text{ in general} \\ \text{Tr}(\rho^2) &< 1\end{aligned}$$

Both are Hermitian, positive semidefinite, and have trace 1.

Figure 8. Pure and mixed density matrices share the same basic structural constraints.

Pure states

$$\rho = |\psi\rangle\langle\psi|, \rho^2=\rho, \text{Tr}(\rho^2)=1.$$

Mixed states

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \text{ with } p_k \geq 0 \text{ and } \sum_k p_k=1.$$

For a mixed state, ρ^2 usually differs from ρ and $\text{Tr}(\rho^2)<1$. The quantity $\text{Tr}(\rho^2)$ is called purity.

Measurement and expectation

$$p(k)=\text{Tr}(\rho P_k), \text{ \<M\>}=\text{Tr}(\rho M).$$

Density matrices are not generally unitary. They represent states, not reversible transformations.

12. The Pauli matrices

X $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ eigenstates $+\rangle, -\rangle$ <i>Bloch x-axis</i>	Y $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ eigenstates $+i\rangle, -i\rangle$ <i>Bloch y-axis</i>	Z $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ eigenstates $0\rangle, 1\rangle$ <i>Bloch z-axis</i>
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Each Pauli matrix is Hermitian, unitary, normal, traceless, and has eigenvalues +1 and -1.

Figure 9. The three Pauli operators define the principal measurement axes of a qubit.

Definitions

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Shared properties

- X, Y, and Z are Hermitian and unitary.
- Each squares to I and is therefore its own inverse.
- Each is traceless and has determinant -1.
- Each has eigenvalues +1 and -1.
- They are pairwise orthogonal under the Hilbert-Schmidt inner product $\text{Tr}(A^\dagger B)$.

Multiplication relations

$$XY = iZ, YZ = iX, ZX = iY.$$

$$YX = -iZ, ZY = -iX, XZ = -iY.$$

Hence distinct Pauli matrices anticommute:

$$\{X, Y\} = XY + YX = 0, \text{ and similarly for the other pairs.}$$

The set $\{I, X, Y, Z\}$ forms a basis for all 2×2 complex matrices.

13. Hadamard and phase gates

Hadamard gate

$$H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

H is real, symmetric, Hermitian, unitary, normal, and self-inverse. It exchanges the Z and X bases:

$$H|0\rangle = |+\rangle, H|1\rangle = |-\rangle, H|+\rangle = |0\rangle, H|-\rangle = |1\rangle.$$

Phase gates

$$S = \text{diag}(1, i), T = \text{diag}(1, \exp(i\pi/4)).$$

S and T are diagonal and unitary, but not Hermitian. They preserve computational-basis probabilities while changing the relative phase.

Relations

$$S^2 = Z, T^2 = S, T^4 = Z, T^8 = I.$$

Rotation gates

$$R_n(\theta) = \exp(-i\theta (\mathbf{n} \cdot \boldsymbol{\sigma})/2).$$

Here $\mathbf{n}$ is a unit Bloch-sphere axis and $\boldsymbol{\sigma} = (X, Y, Z)$. The generator $\mathbf{n} \cdot \boldsymbol{\sigma}$ is Hermitian; the exponential is unitary.

14. Permutation and controlled matrices

A permutation matrix has exactly one entry 1 in every row and every column, with all other entries zero. It permutes standard basis vectors.

Properties

- Every permutation matrix is unitary and orthogonal.
- Its inverse is its transpose.
- It implements a reversible classical function on computational-basis states when the function is bijective.

Examples

$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is the two-element swap permutation.

CNOT is a 4 x 4 permutation matrix that swaps $|10\rangle$ and $|11\rangle$ while leaving $|00\rangle$ and $|01\rangle$ fixed.

Controlled operations as block matrices

$$\text{Controlled-U} = \begin{array}{|c|c|} \hline \mathbf{I} & 0 \\ \hline 0 & \mathbf{U} \\ \hline \end{array}$$

control = 0: apply I

control = 1: apply U

Block matrices encode conditional quantum operations compactly.

Figure 10. A controlled-U gate is block diagonal in the control-qubit basis.

$$C(U) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U.$$

If U is unitary, then $C(U)$ is also unitary. The block structure exposes why: each orthogonal control subspace receives a unitary operation.

15. Tensor-product structure

Composite quantum systems use Kronecker products. If A acts on system A and B on system B , the joint operator is A tensor B .

Adjoint and multiplication rules

$$(A \text{ tensor } B)^\dagger = A^\dagger \text{ tensor } B^\dagger.$$

$$(A \text{ tensor } B)(C \text{ tensor } D) = AC \text{ tensor } BD.$$

Class preservation

- If A and B are unitary, A tensor B is unitary.
- If A and B are Hermitian, A tensor B is Hermitian.
- If A and B are positive semidefinite, A tensor B is positive semidefinite.
- If P and Q are projectors, P tensor Q is a projector.

Eigenvalues

If $A|a\rangle = \alpha|a\rangle$ and $B|b\rangle = \beta|b\rangle$, then

$$(A \text{ tensor } B)(|a\rangle \text{ tensor } |b\rangle) = \alpha\beta (|a\rangle \text{ tensor } |b\rangle).$$

Local gates

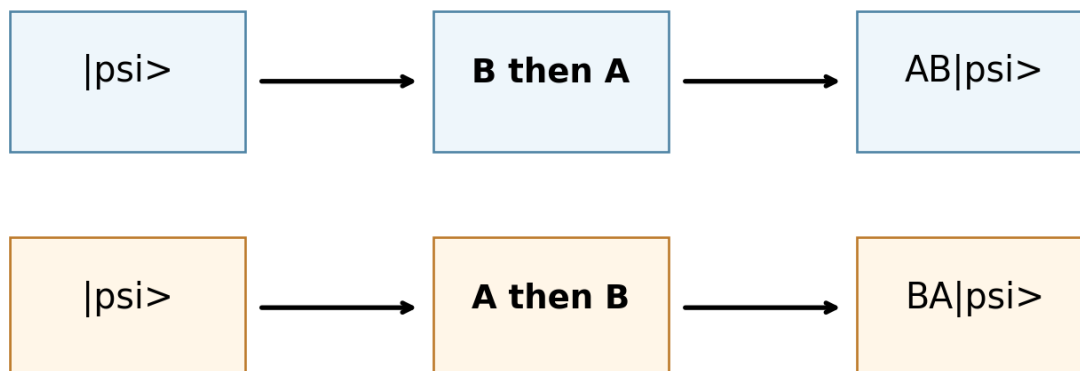
U on the first qubit of a two-qubit register: U tensor I .

U on the second qubit: I tensor U .

Always match the tensor-factor ordering to the convention used by the text or software library.

16. Commutators and anticommutators

$$[A,B]=AB-BA, \{A,B\}=AB+BA.$$



$[A,B] = AB - BA$ measures the failure of operation order to commute.

Figure 11. The commutator measures the difference between the two operation orders.

Commuting Hermitian observables

If $[A,B]=0$, finite-dimensional Hermitian A and B possess a common orthonormal eigenbasis. Their values can be simultaneously definite.

Noncommuting observables

Nonzero commutators underlie uncertainty and basis incompatibility. For Pauli matrices:

$$[X,Y]=2iZ, [Y,Z]=2iX, [Z,X]=2iY.$$

Anticommutators

$$\{X,Y\}=\{Y,Z\}=\{Z,X\}=0.$$

For unit vectors a and b ,

$$(a \cdot \sigma)(b \cdot \sigma) = (a \cdot b)I + i(a \times b) \cdot \sigma.$$

This identity packages both the commutator and anticommutator structure of qubit observables.

17. Matrix exponentials and generators

The exponential of a matrix is defined by its power series:

$$\exp(A) = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

If A is diagonalizable, exponentiation acts on its eigenvalues. If $A = VDV^{-1}$, then

$$\exp(A) = V \exp(D) V^{-1}.$$

Hermitian generator to unitary evolution

$$H^\dagger = H \Rightarrow U(t) = \exp(-iHt) \text{ is unitary.}$$

Exponentiating a Hermitian generator produces a unitary family

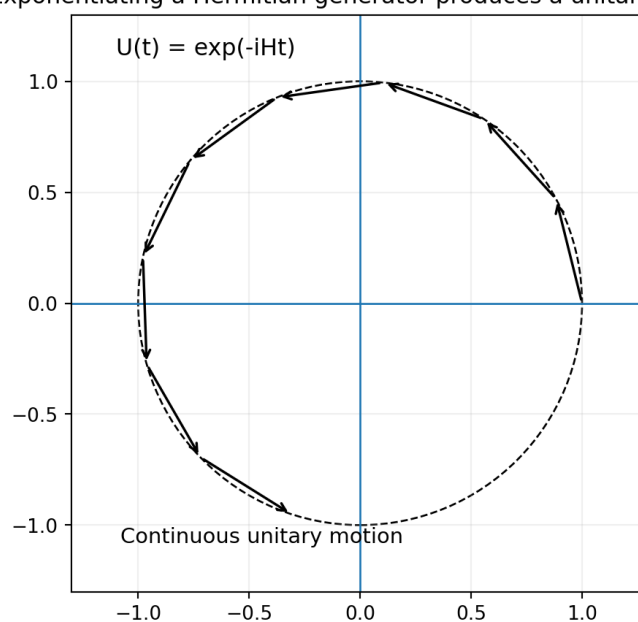


Figure 12. Continuous unitary motion is generated by a Hermitian operator.

Proof sketch

Because $(-iHt)^\dagger = iHt$, we have $U(t)^\dagger = \exp(iHt)$. Since H commutes with itself,

$$U^\dagger U = \exp(iHt)\exp(-iHt) = I.$$

Pauli exponential identity

Since $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$,

$$\exp(-i\theta \mathbf{n} \cdot \boldsymbol{\sigma}) = \cos(\theta/2)I - i \sin(\theta/2)\mathbf{n} \cdot \boldsymbol{\sigma}.$$

This is the algebraic foundation of single-qubit rotations.

18. Worked classification examples

Example A: $A = \text{diag}(2, 3)$

A is diagonal, Hermitian, normal, and positive definite. It is not unitary because $A^\dagger A = \text{diag}(4, 9)$, not I.

Example B: $S = \text{diag}(1, i)$

S is diagonal, unitary, and normal. It is not Hermitian because $S^\dagger = \text{diag}(1, -i)$ differs from S. It is not positive semidefinite because positive semidefinite matrices must be Hermitian.

Example C: $P_0 = |0\rangle\langle 0|$

P_0 is diagonal, Hermitian, normal, positive semidefinite, and a projector. It is not unitary because it maps $|1\rangle$ to zero and is not invertible.

Example D: X

X is a permutation matrix, Hermitian, unitary, normal, and self-inverse. It is not positive semidefinite because one eigenvalue is -1.

Example E: density matrix

$$\rho = (1/2)[[1, r], [r^*, 1]], \text{ with } |r| \leq 1.$$

ρ is Hermitian, positive semidefinite, and trace one. It is pure only when $|r|=1$; otherwise it is mixed. It is generally neither unitary nor a projector.

Example F: non-normal matrix

$$J = [[1, 1], [0, 1]].$$

J is neither Hermitian nor unitary and satisfies $JJ^\dagger \neq J^\dagger J$. It is defective and has no orthonormal eigenbasis.

19. Common mistakes and diagnostic checks

Mistake	Correction
Using transpose instead of adjoint	Conjugate complex entries before transposing.
Assuming Hermitian means unitary	Hermitian requires $A^\dagger = A$; unitary requires $A^\dagger A = I$.
Assuming every observable is a gate	Observables need not preserve norm or be reversible.
Assuming every projector is unitary	A nontrivial projector loses components and is noninvertible.
Believing unitary eigenvalues are real	They only need magnitude one; phases such as i are allowed.
Calling a non-Hermitian matrix positive	Positive semidefinite matrices are Hermitian by the standard definition.
Thinking normal means diagonal	Normal means unitarily diagonalizable, possibly after changing basis.
Treating density matrices as state vectors	A density matrix is an operator with trace one, not a ket.
Ignoring global phase in controlled contexts	A phase global on U may become relative in controlled- U .
Assuming $AB=BA$	Check the commutator; quantum operation order often matters.

Fast classification checklist

- 1 Compute $A^\dagger$.
- 2 Check $A^\dagger = A$ for Hermitian.
- 3 Check $A^\dagger A = I$ for unitary.
- 4 Check $AA^\dagger = A^\dagger A$ for normal.
- 5 Check $A^2 = A$ for projector.
- 6 For Hermitian A , inspect eigenvalues to test positivity.
- 7 Check trace, determinant, rank, and invertibility as supporting diagnostics.

20. Exercises

A. Classification

- 1 Classify I , 0 , X , Z , H , S , P_0 , and $\text{diag}(2,3)$ as Hermitian, unitary, normal, projector, and positive semidefinite where applicable.
- 2 Determine whether $A = \begin{bmatrix} 2 & 1+i \\ 1-i & 4 \end{bmatrix}$ is Hermitian and positive definite.
- 3 Determine whether $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}$ is unitary.
- 4 Give an example of a unitary matrix that is not Hermitian and a Hermitian matrix that is not unitary.
- 5 Find the Hermitian and skew-Hermitian parts of $A = \begin{bmatrix} 1 & i \\ i & 2, 3-i \end{bmatrix}$.

B. Projectors and positivity

- 1 Compute $|+\rangle\langle+|$ and verify $P^2 = P$.
- 2 For $|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle$, calculate $\langle\psi|P_0|\psi\rangle$.
- 3 Show that $B^\dagger B$ is positive semidefinite for every B .
- 4 Find the eigenvalues of $\begin{bmatrix} 1 & r \\ r & 1 \end{bmatrix}$ and determine when it is positive semidefinite.
- 5 Prove that a nonzero projector cannot be negative definite.

C. Pauli and commutator algebra

- 1 Verify $XY = iZ$ and $YX = -iZ$.
- 2 Compute $[X, Z]$ and $\{X, Z\}$.
- 3 Show that $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$ for any real unit vector $\mathbf{n}$.
- 4 Use the previous result to derive $\exp(-i\theta \mathbf{n} \cdot \boldsymbol{\sigma}/2)$.
- 5 Express the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ as a linear combination of I, X, Y, Z .

D. Composite and advanced structures

- 1 Prove that $A \otimes B$ is unitary when A and B are unitary.
- 2 Write the 4×4 matrix of controlled- Z and classify it.
- 3 Show that $\rho = I/2$ is Hermitian, positive semidefinite, trace one, and maximally mixed.
- 4 For $\rho = \text{diag}(p, 1-p)$, determine its purity $\text{Tr}(\rho^2)$ and the values of p for which it is pure.
- 5 Explain why $\exp(-iHt)$ is unitary when H is Hermitian.

21. Selected solutions

Exercise A2

A is Hermitian because the diagonal entries are real and the off-diagonal entries are conjugates. Its eigenvalues are 3 plus or minus $\sqrt{3}$, both positive, so A is positive definite.

Exercise A3

$U^\dagger = (1/\sqrt{2})[[1, -i], [-i, 1]]$. Multiplication gives $U^\dagger U = I$, so U is unitary. It is not Hermitian.

Exercise A5

$$H_A = (A + A^\dagger)/2, K_A = (A - A^\dagger)/2.$$

Substitute the matrix and simplify entry by entry. H_A must satisfy $H_A^\dagger = H_A$, while $K_A^\dagger = -K_A$. Their sum reproduces A.

Exercise B1

$$|+\rangle\langle+| = (1/2)[[1, 1], [1, 1]].$$

Squaring gives the same matrix because each row-column product produces one half after the prefactor is included.

Exercise B3

$$\langle v|B^\dagger B|v\rangle = \langle Bv|Bv\rangle = \|Bv\|^2 \geq 0.$$

Therefore $B^\dagger B$ is positive semidefinite.

Exercise C2

$$[X, Z] = -2iY, \{X, Z\} = 0.$$

The zero anticommutator expresses the fact that X and Z correspond to perpendicular Bloch axes.

Exercise C5

The coefficients are

$$\alpha = (a+d)/2, \beta = (b+c)/2, \gamma = (c-b)/(2i), \delta = (a-d)/2,$$

so $A = \alpha I + \beta X + \gamma Y + \delta Z$.

Exercise D2

$$CZ = \text{diag}(1, 1, 1, -1).$$

CZ is diagonal, Hermitian, unitary, normal, self-inverse, and has determinant -1. It is not positive semidefinite because it has eigenvalue -1.

Exercise D4

$$\text{Tr}(\rho^2) = p^2 + (1-p)^2.$$

Purity equals 1 only at $p=0$ or $p=1$; otherwise the state is mixed. It reaches its minimum $1/2$ at $p=1/2$.

22. Compact reference sheet

Class	Condition	Key consequence
Hermitian	$A^\dagger = A$	Real eigenvalues; observable
Skew-Hermitian	$A^\dagger = -A$	Purely imaginary eigenvalues; unitary generator
Unitary	$U^\dagger U = I$	Preserves inner products; gate
Orthogonal	$Q^T Q = I$, Q real	Real unitary transformation
Normal	$AA^\dagger = A^\dagger A$	Orthonormal eigenbasis
Projector	$P^2 = P$; orthogonal if $P^\dagger = P$	Selects a subspace
Positive semidefinite	A Hermitian and $\langle v A v \rangle \geq 0$	Nonnegative eigenvalues
Density matrix	$\rho^\dagger = \rho$, $\rho \geq 0$, $\text{Tr } \rho = 1$	Quantum state operator
Special unitary	U unitary, $\det U = 1$	Rotation without determinant phase

Essential formulas

$(AB)^\dagger = B^\dagger A^\dagger$
$H = \sum_k \lambda_k v_k\rangle\langle v_k $
$p_k = \langle \psi P_k \psi \rangle$
$U(t) = \exp(-iHt)$
$[A, B] = AB - BA$, $\{A, B\} = AB + BA$
$R_n(\theta) = \cos(\theta/2)I - i \sin(\theta/2) \mathbf{n} \cdot \boldsymbol{\sigma}$

Recognition rules

- Real eigenvalues suggest Hermitian structure, but verify the adjoint condition.
- Eigenvalues on the unit circle suggest unitary structure, but normality and a complete eigenbasis matter.
- Eigenvalues 0 or 1 suggest a projector when the matrix is diagonalizable; verify $P^2 = P$.
- Nonnegative eigenvalues characterize positive semidefinite Hermitian matrices.
- Orthonormal columns are the fastest practical unitarity test.

End of Topic 6

The next foundation topic is basic probability for quantum measurement: random variables, conditional probability, expectation, variance, sampling, and experimental shot counts.